

SANDY BROWN

Consultants in Acoustics, Noise & Vibration

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Greenwich Millennium Village, Phases 3, 4 & 5 Buildings 302, 303 & 304

Planning condition 87: Internal and facade sound levels

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A	14 Mar 18		Gareth Wells	Jason Swan
B	11 Apr 18	Building elevations included	Gareth Wells	Jason Swan
C	17 Apr 18	Cited reports updated	Gareth Wells	Jason Swan

Summary

Sandy Brown Associates LLP (Sandy Brown) has been appointed to provide acoustic assessments and design strategy advice in relation to the residential master plan for Greenwich Millennium Village, Phases 3, 4 and 5, along with more detailed construction advice for the latest sub-phase including Buildings 302, 303 & 304.

This document outlines the strategy to be employed to comply with planning condition 87 of the outline planning permission 14/1633/MA (which amended the original consent 12/0022/O) relating to facade and internal noise levels imposed by the Royal Borough of Greenwich (RBG) for Buildings 302, 303 & 304.

An overview of the acoustic criteria is included along with details of the overall acoustic design related to the condition. Details are then provided for the strategy to mitigate noise where areas of the facade marginally exceed the criteria.

The external and internal noise level criteria are presented, referencing the six criteria from the Environmental Statement, in line with planning condition 87.

- Criteria 1 and 2 are considered using acoustic models to determine the facade noise level from the various wharf and dredger noise sources.
- Facade design details are then developed to meet Criteria 3, 4, 5 and 6.

A typical worked example is included to show how the design is developed for specific apartments.

The acoustic design strategy set out in this report, which has been implemented by Jestico + Whiles, complies with the requirements of planning condition 87.

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1 Introduction

Sandy Brown Associates LLP (Sandy Brown) has been appointed to provide acoustic assessments and design strategy advice in relation to the residential master plan for Greenwich Millennium Village, Phases 3, 4 and 5, along with more detailed construction advice for the latest sub-phase including Buildings 302, 303 & 304.

This document sets out the strategy that is to be adopted in order to allow the development to comply with condition 87 of the outline planning permission 14/1633/MA (which amended the original consent 12/0022/O) as granted by the Royal Borough of Greenwich.

The condition is worded as follows.

Condition 87:

‘Prior to first occupation of the residential units in Parcel 1 and each subsequent relevant phase or part of the development, full details demonstrating that criteria 1-6 in paragraph 15.93 of the Environmental Statement are met and also that the internal noise level for all non-habitable rooms within the development meets LAeq45 (dB) during the day and LAeq35 (dB) during the night, shall be submitted to and approved in writing by the Local Planning Authority (in consultation with the PLA and Wharf operators).’

Though the acoustic design has been developed in detail for earlier parcels (ie, Buildings 101 to 109, 206 to 210), the strategies employed are applicable to the whole masterplan.

This document should be read alongside the following documents to fully understand how the acoustic requirements are implemented:

- Latest architectural drawings;
- Sandy Brown’s report 17470-R01-A *Environmental noise survey report* 14 March 2018;
- Sandy Brown’s report 17470-R04-B *Building 302 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- Sandy Brown’s report 17470-R05-B *Building 303 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- Sandy Brown’s report 17470-R06-B *Building 304 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;

The revision of 17470-R04, R05 & R06 given above is the latest at the time the present report was written. Should the above reports be revised, the references given herein automatically apply to the latest version.

2 Internal noise level criteria for residences

The six criteria from the Environmental Statement referenced in the planning condition are reproduced below:

- **Criterion 1** is the maximum external facade noise level from all wharf and dredger sources operating at a cumulative maximum, for habitable rooms with opening windows ie, L_{Aeq} 44 dB.
 - Where facade levels exceed L_{Aeq} 44 dB, habitable room windows will be fixed shut.¹
 - Non-habitable room windows would only open if it can be demonstrated that noise levels in the nearest habitable rooms are $\leq L_{Aeq}$ 30 dB with all internal doors open.
- **Criterion 2** is the maximum dredgers-only noise level at 63 Hz octave-band, inside any habitable room i.e. $L_{A,63Hz}$ 24 dB (that is $L_{F,63Hz}$ 50 dB or NR 19).²
 - This will ensure a level of extra protection to residents from low-frequency noise in particular where facade levels exceed 39 dB(A) at 63 Hz, i.e. habitable room windows are recommended to be fixed shut.¹
 - Wall / ceilings affected by excess low-frequency noise should have an enhanced construction or insulation.
- **Criterion 3** is the maximum noise level in habitable rooms at night ie, L_{Aeq} 30 dB [WHO, BS 8233:1999 ‘good’]
 - Includes all external noise sources (i.e. Criterion 1 plus night-time traffic / environmental sources).
 - To include mechanical services noise.
- **Criterion 4** is the maximum noise level in bedrooms during the day ie, L_{Aeq} 35 dB [BS 8233:1999 ‘reasonable’]. The maximum noise level in other habitable rooms during the day is L_{Aeq} 40 dB [BS 8233:1999 ‘reasonable’].
 - Includes all external noise sources (ie, Criterion 1 plus daytime traffic / environmental sources).
 - To include mechanical services noise.
- **Criterion 5** is the maximum target noise level in public / private outdoor areas (ie, balconies, gardens) ie, L_{Aeq} 55 dB [BS 8233:1999 ‘upper limit’].
 - Includes all external noise sources (ie, Criterion 1 plus traffic / environmental sources).
- **Criterion 6** is that individual noise events shall not normally exceed L_{AFmax} 45 dB in bedrooms at night [BS8233:1999].

¹ Localised screening is used to achieve this in some areas.

² To clarify, Criterion 2 refers to the L_{Aeq} equivalent continuous noise level.

Criteria 1 and 2 are the first to consider as they determine whether windows can be opened.

The apartments in Buildings 302, 303, & 304 are understood will all be mechanically ventilated. Therefore the internal noise levels required by Criteria 3, 4 and 6 can be met with windows closed and with the minimum sound insulation performances set out in reports 17470-R04, R05 & R06 in place.

Extensive 3D computer noise modelling has been carried out using the software CadnaA to determine the facade noise levels for each building within the development.

3 External sound insulation (building envelope / ventilation strategy)

For the purposes of this report, 'building envelope' or 'cladding' shall refer to the facade, windows, roof, roof lights, ventilation openings, and any other external envelope element of the buildings.

3.1 Facade noise levels

Section 2 of this report outlines the internal noise levels required under various site conditions and the facade noise limits governing whether windows are to be fixed shut.

To assess what will be needed to meet these criteria, extensive environmental and source noise investigations have been undertaken at the site by Sandy Brown Associates, Hoare Lea and others.

This data has been incorporated into computer models using the software program CadnaA. The models have been run under the various conditions for each of the Criteria and their results and implications are shown in the numerous colour-coded figures following this page, and are described below.

Since buildings within the development provide acoustic screening to other buildings (ie, Buildings 103 to 107 and 210 provide screening to the lower buildings), and other future buildings may reflect noise back on to the site, the noise levels incident on each facade will differ at various stages of development.

The intention is to occupy each building on its completion as the scheme progresses. In light of this, it is necessary to consider the noise levels for each facade at the point it will be occupied and also at the completion of the master plan, and design to the worst case (ie, the highest noise level which might occur).

The following pages show the results of the computer modelling for 'at occupation' condition, and then the entire masterplan. These are indicated in Figure 1 and Figure 2, respectively. As this report focusses on Buildings 302, 303 & 304, only the results for these buildings are shown.

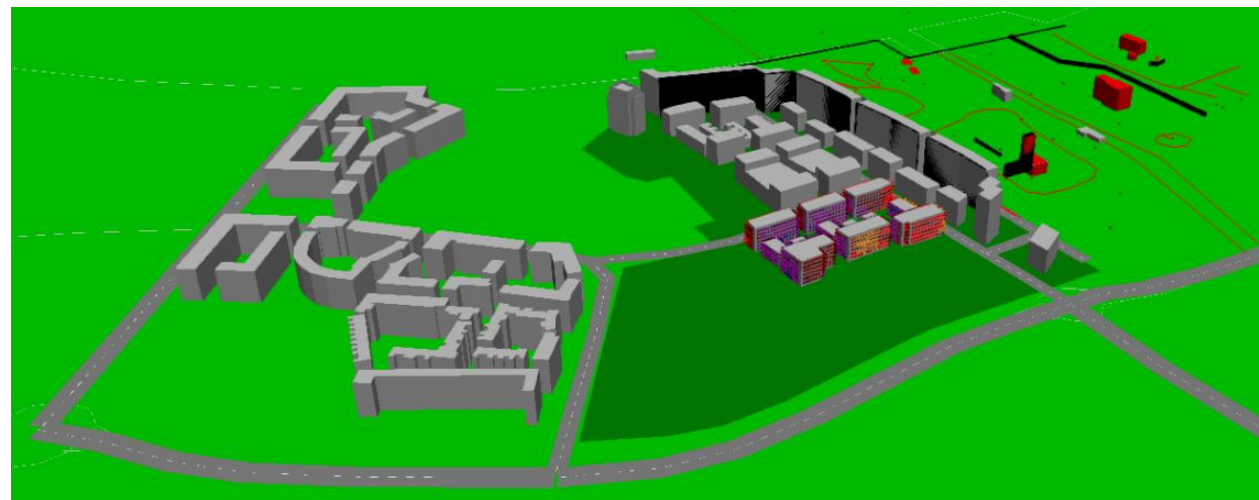


Figure 1 Buildings included in Parcel 1 and Parcel 2 sub-phase 1 model

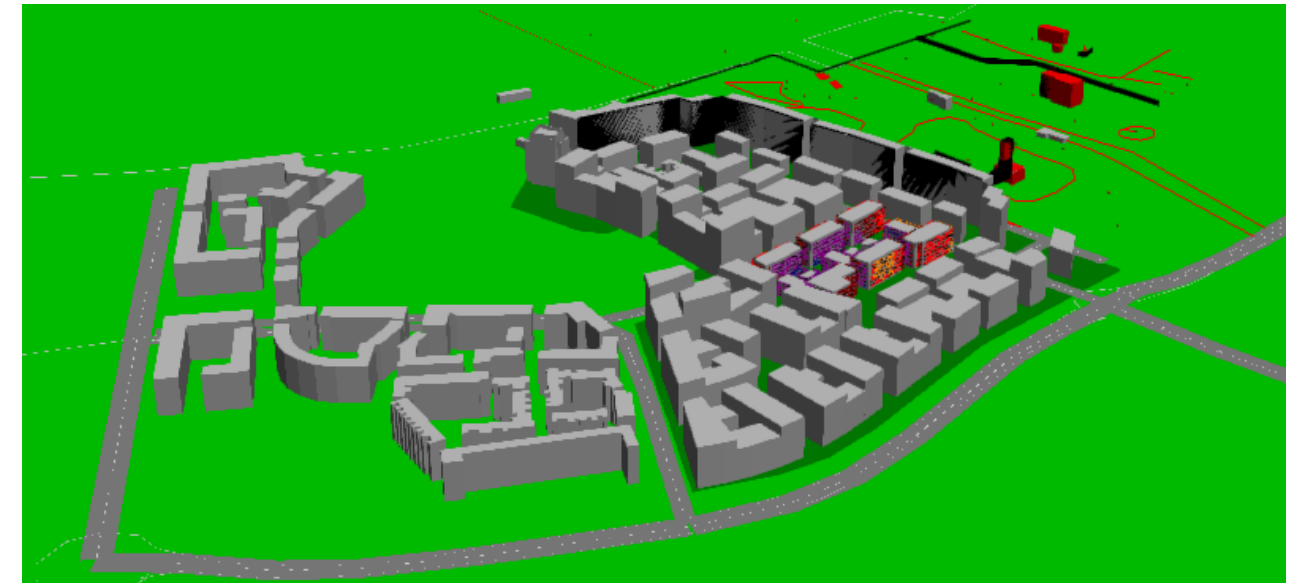


Figure 2 Buildings included in Masterplan model

The following sections refer to Figure 3 to Figure 18 inclusive with the following distribution:

- Figure 3 to Figure 10 show the noise levels at the facades of Buildings 302, 303 & 304 with only Buildings 101 to 109, 206 to 210, and 504 in place, the 'at occupation' condition;
- Figure 11 to Figure 18 show the noise levels at the facades of Buildings 302, 303 & 304 with the whole masterplan in place.

3.1.1 Figure 3 to Figure 4 and Figure 11 to Figure 12 – Criterion 1, maximum broadband noise from 24-hour wharf operation

The noise coming specifically from operations associated with the Angerstein and Murphy's wharves including the dredgers, aggregate works, and all other related sources are to be assessed on their own to better ensure minimal complaint specifically from their operation.

Taking the unlikely but conservative condition that all wharf / dredger sources operate at once, these figures show the resulting noise level at the facade of each residence. Where noise levels are above L_{Aeq} 44 dB (navy blue and worse), Criterion 1 requires windows of habitable rooms to be fixed closed.

Non-habitable rooms (eg, bathrooms, small separate kitchens, and foyers) will be assessed on their specific configuration in relation to habitable rooms through open doors, but generally will have their windows fixed closed if facade noise levels from wharf and dredger sources are above L_{Aeq} 50 dB.

Given the existing noise environment on site with significant contributions from road traffic and aircraft to and from London City Airport, it is worth noting that it will not be possible to measure the noise levels solely from the operation of the wharf and dredgers on site, nor could they ever run all processes (at maximum) simultaneously. As a result, Criterion 1 can only be shown through computer modelling.

3.1.2 Figure 5 to Figure 6 and Figure 13 to Figure 14 – Criterion 2, low frequency noise from dredgers

The noise associated with the dredger vessels anchored at Angerstein and Murphy's Wharves is important to consider on its own, as its low-frequency content makes it prone to causing complaint, especially since it must be free to occur at any time of the day or night due to tidal considerations. As such, Criterion 2 aims to control the noise in the 63 Hz octave band to no more than $L_{Aeq,63Hz}$ 24 dB inside any habitable room.

The figures show the broadband facade noise level at Buildings 302, 303 & 304 resulting from a dredger docked at both wharves with the buildings in place at Buildings' 302, 303 & 304 occupation (Figure 5 to Figure 6), and at the completion of the masterplan (Figure 13 to Figure 14). As the dredger noise spectrum is dominated by noise in the 63 Hz octave-band, it can be shown that:

- For facades of buildings in the acoustic shadow of the north-eastern terrace buildings (Buildings 103 to 107 and 210), which includes all of Buildings 302, 303 & 304, due to the effects of barrier attenuation, the A-weighted broadband facade noise level shown is the same as the A-weighted facade level in the 63 Hz octave-band alone.

Table 1 shows the resulting facade sound insulation requirement needed to cover noise at 63 Hz. For a window in an affected facade to open partially (ie, < 1% of a room's facade area), the external facade noise level should be no higher than $L_{Aeq,63Hz}$ 39 dB. This is based on available data that at 63 Hz the sound insulation of a partial opening would comfortably be no less than $D_{n,e}$ 15 dB.

3.1.3 Figure 7 to Figure 8 and Figure 15 to Figure 16 – Criterion 3 (and 6), night-time noise sources

The first two criteria deal with specific noise sources unique to this site and their intrinsic concern with complaint. Other more common noise sources will also affect the site such as road traffic and aircraft. When these are combined with the noise sources from the wharf and dredgers, there can be a better understanding of the total environmental noise conditions at the site.

The figures show the facade noise levels for Buildings 302, 303 & 304 expected during the night-time hours from 23:00 to 07:00 with the buildings existing at the occupation of Buildings 302, 303 & 304

(Figure 7 to Figure 8) and at the completion of the masterplan (Figure 15 to Figure 16). These include the 100% noise from the wharf / dredger operations as well as traffic noise from night-time counts provided to Sandy Brown as part of the EIA and Sandy Brown's own measurements on site.

Based on the facade noise levels shown in these figures, the sound insulation and ventilation strategy can be determined for bedrooms, living rooms, and other rooms so that Criterion 3 (L_{Aeq} 30 dB in habitable rooms) can be achieved inside the residences at night.

The specification key for facade sound insulation follows these figures in Table 2.

Note that these sound insulation performances may need to be increased where maximum internal noise levels in bedrooms might regularly exceed L_{AFmax} 45 dB. Maximum noise levels are expected to decrease for most of the site in the acoustic shadow of the eastern terrace, but this may still be an issue where bedroom facades have view of Bugsby's Way or the aggregate works.

3.1.4 Figure 9 to Figure 10 and Figure 17 to Figure 18 – Criteria 4 & 5, Commercial/Retail, daytime noise sources

Similarly, these figures show the facade noise levels during the daytime hours from 07:00 to 23:00. These include noise from the wharf / dredger operations, the waste management plant, and traffic noise from daytime traffic counts provided to Sandy Brown as part of the EIA and Sandy Brown's own measurements on site.

Based on the facade noise levels shown in these figures, the sound insulation and ventilation strategy can be determined for bedrooms, living rooms, and other rooms so that Criteria 4 (L_{Aeq} 35 dB in bedrooms, L_{Aeq} 40 dB in other habitable rooms) can be achieved inside the residences during the day.

The specification key for facade sound insulation follows these figures in Table 3.

For Criterion 5, these figures can be used to determine if an outdoor area such as a balcony will have noise levels at or below L_{Aeq} 55 dB by looking at the highest nearby facade noise level.

‘At occupied condition’ computer model
CRITERION 1: Wharf/Dredger noise sources only, A-weighted broadband

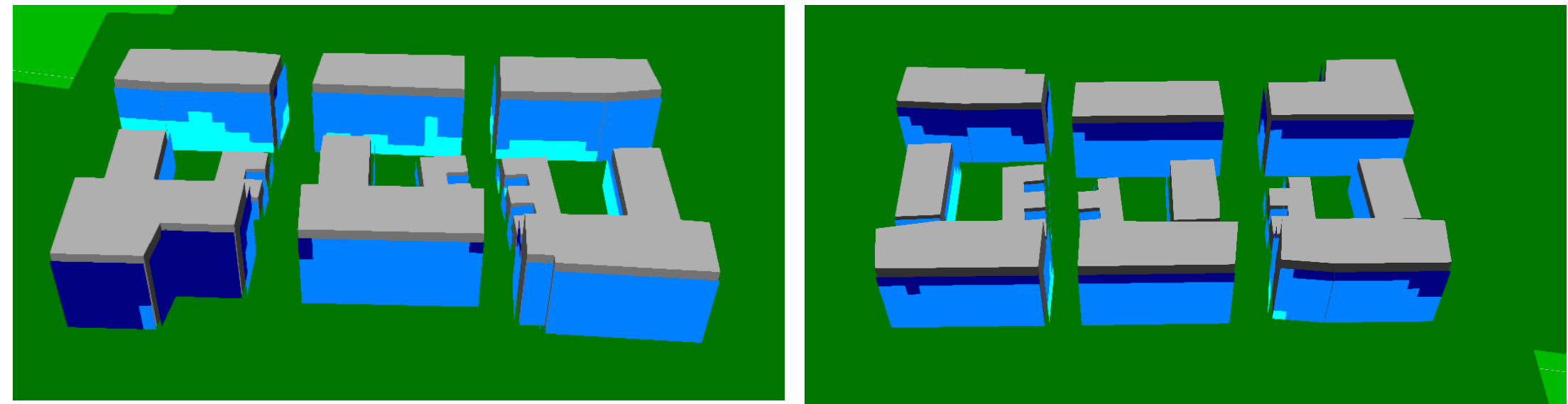


Figure 3 ‘At occupation condition’ model, Criterion 1 - Looking north (left) and south (right)

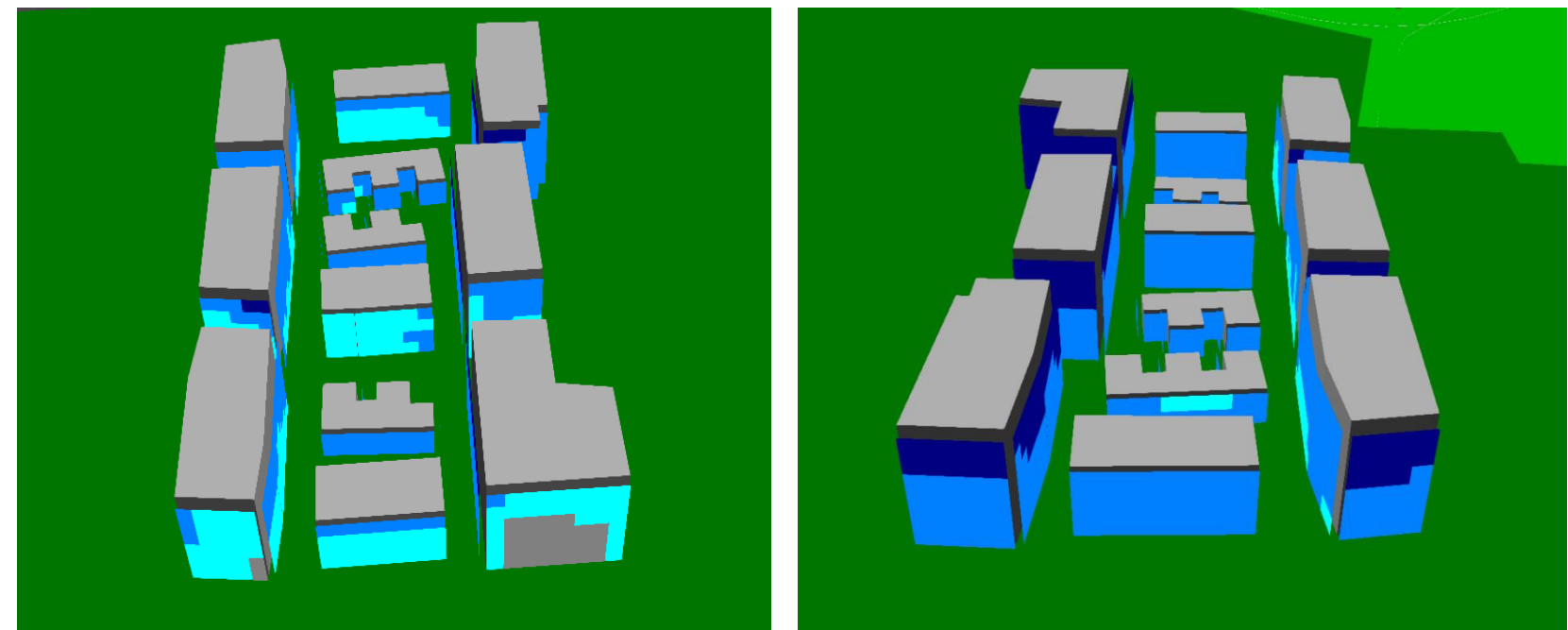


Figure 4 ‘At occupation condition’ model, Criterion 1 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Requirement
Grey	 ≤ 34	Windows can open
Cyan	 35-39	Windows can open
Blue	 40-44	Windows can open
Navy	 45-49	Habitable room windows fixed shut*
Purple	 50-54	Habitable room windows fixed shut*
Maroon	 55-59	Habitable room windows fixed shut

* Unless localised screening is provided.

‘At occupied condition’ computer model

CRITERION 2: Dredger noise sources only, A-weighted broadband, 63 Hz dominates

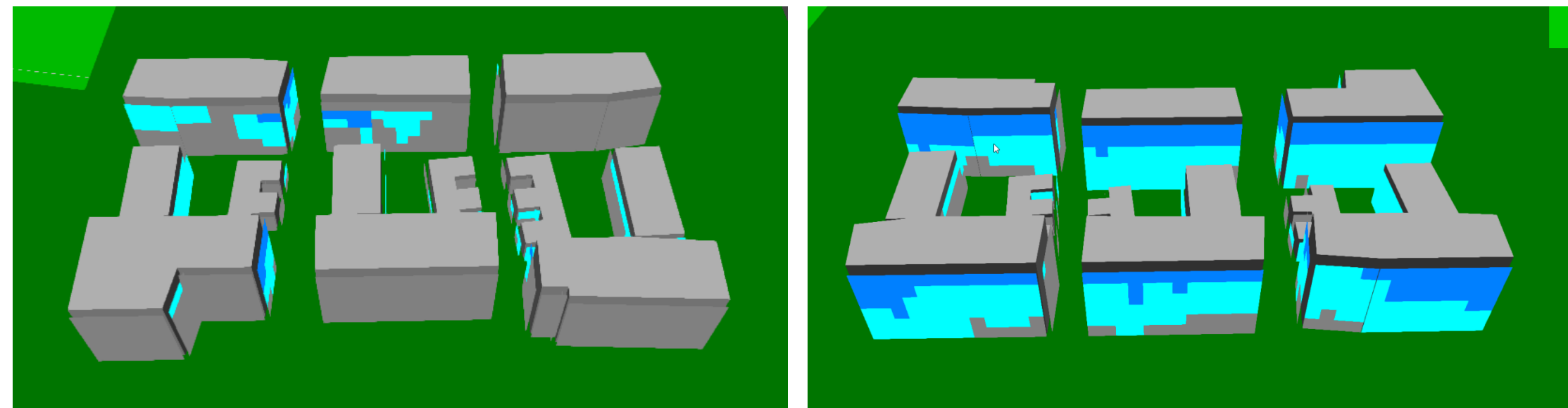


Figure 5 ‘At occupation condition’ model, Criterion 2 - Looking north (left) and south (right)

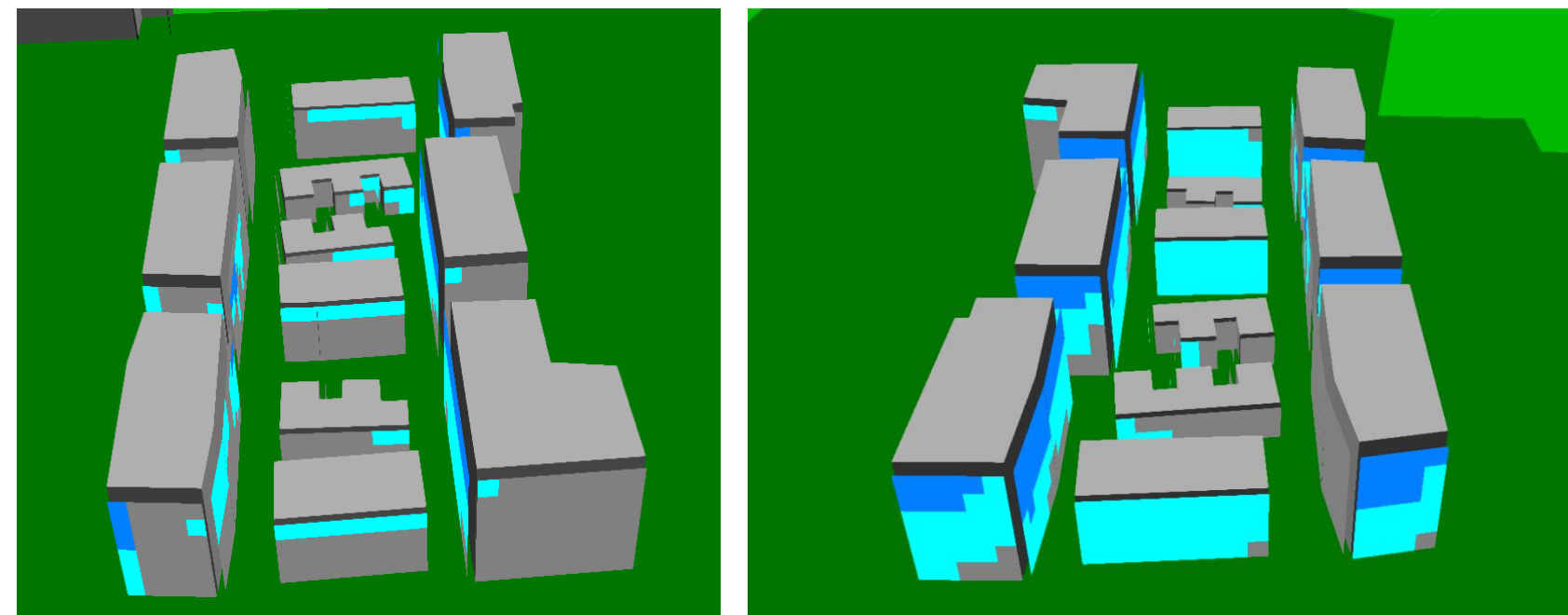


Figure 6 ‘At occupation condition’ model, Criterion 2 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Window Recommendation (See Table 1 for sound insulation)
Grey	 ≤ 34	Windows can open if Criterion 1 met
Cyan	 35-39	Windows open partially if Criterion 1 met
Blue	 40-44	Habitable room windows fixed shut*
Navy	 45-49	All windows fixed shut*
Purple	 50-54	All windows fixed shut
Maroon	 55-59	All windows fixed shut

* Unless localised screening is provided.

‘At occupied condition’ computer model
CRITERION 3: All noise sources, NIGHT, A-weighted broadband facade noise levels

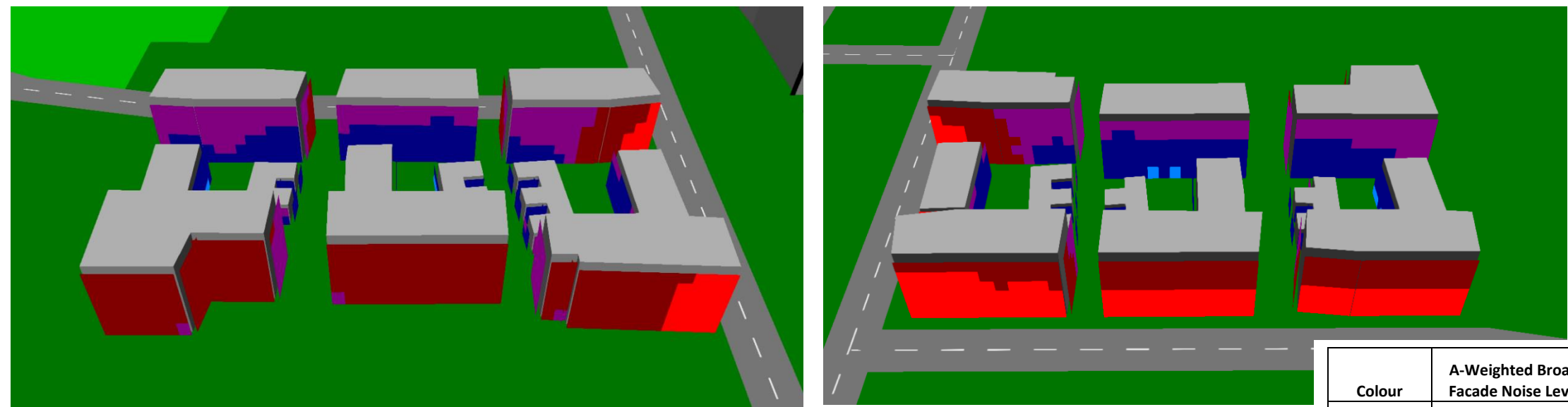


Figure 7 ‘At occupation condition’ model, Criterion 3 - Looking north (left) and south (right)

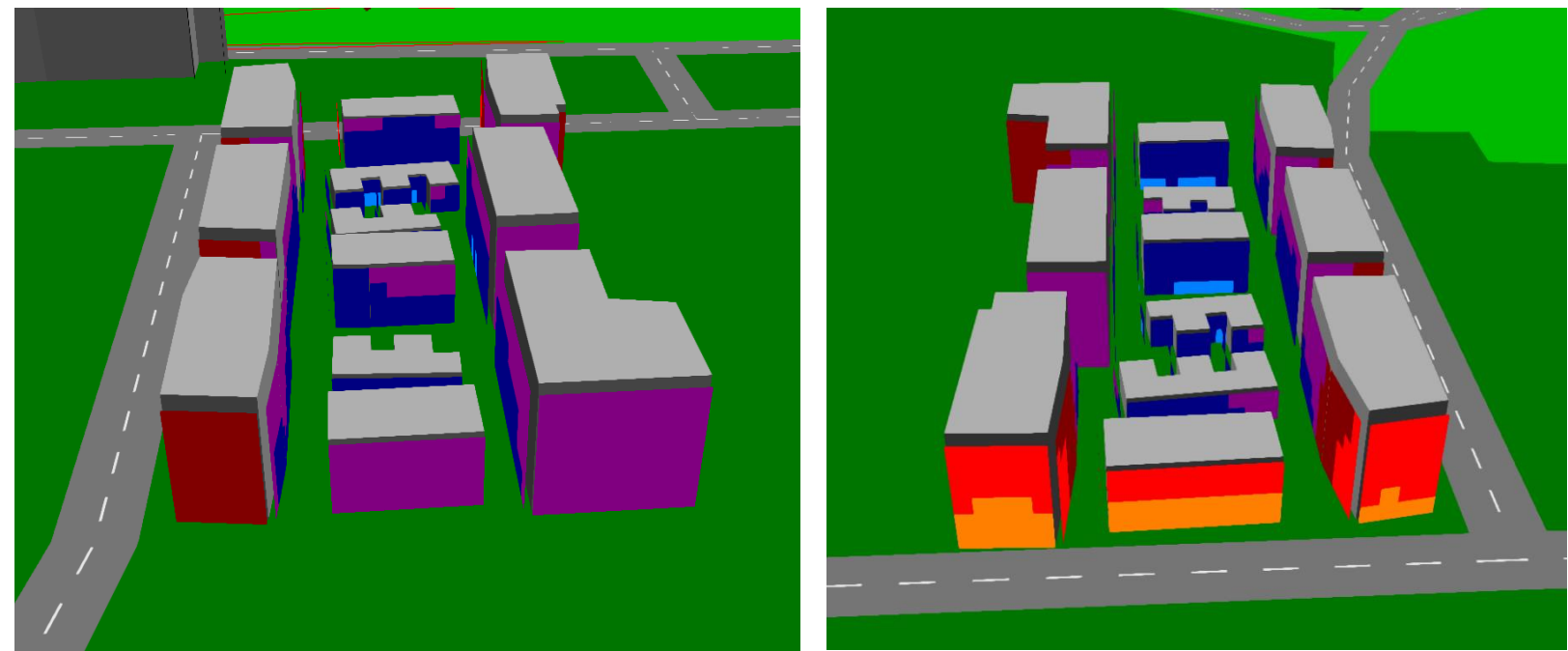


Figure 8 ‘At occupation condition’ model, Criterion 3 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Recommendation
Grey	 ≤ 34	(see Table 2)
Cyan	 35-39	(see Table 2)
Blue	 40-44	(see Table 2)
Navy	 45-49	(see Table 2)
Purple	 50-54	(see Table 2)
Maroon	 55-59	(see Table 2)
Red	 60-64	(see Table 2)
Orange	 65-69	(see Table 2)
Yellow	 70-74	(see Table 2)
Lime	 75-79	(see Table 2)
Green	 80-84	(see Table 2)
Emerald	 85-89	(see Table 2)

‘At occupation condition’ computer model

CRITERIA 4 & 5: All noise sources, DAY, A-weighted broadband facade noise levels

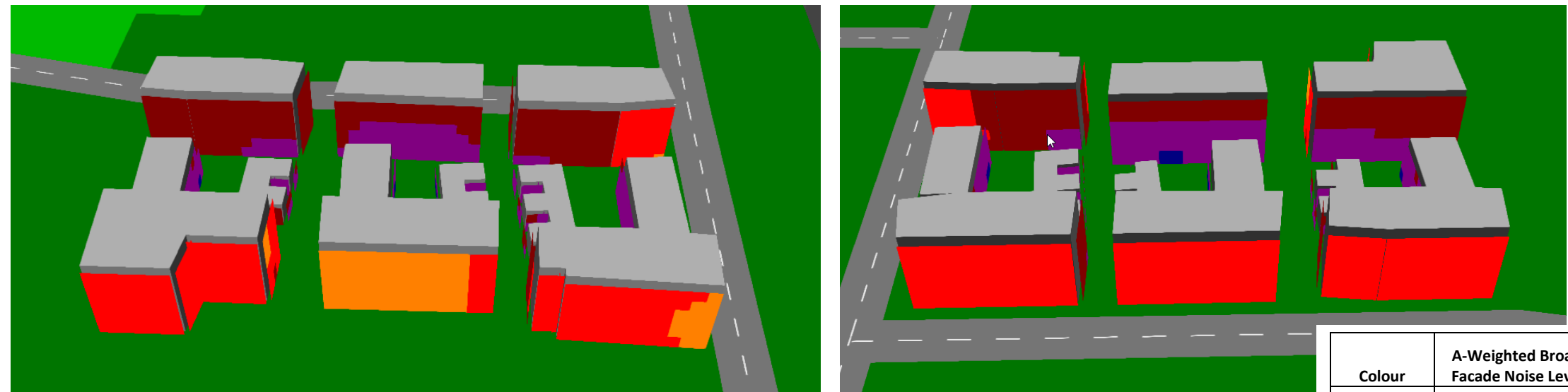


Figure 9 ‘At occupation condition’ model, Criteria 4 & 5 - Looking north (left) and south (right)

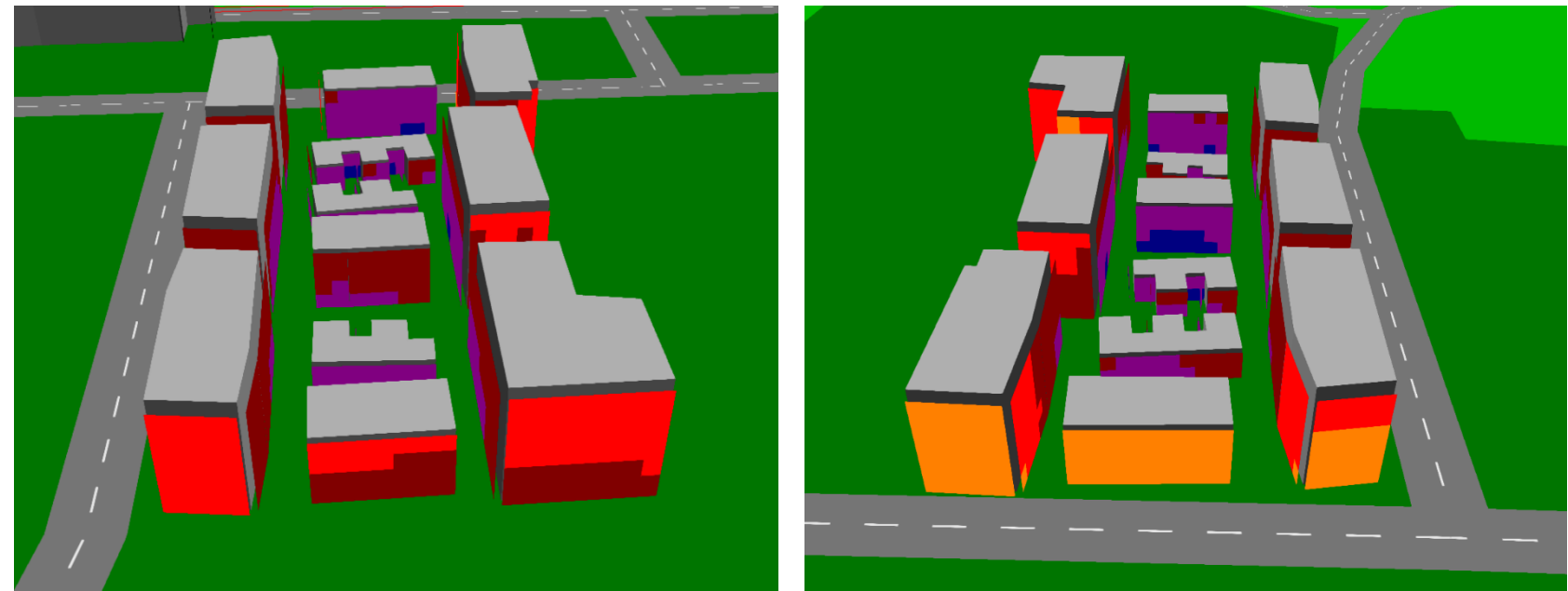


Figure 10 ‘At occupation condition’ model, Criteria 4 & 5 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Recommendation
Grey	 ≤ 34	(see Table 3)
Cyan	 35-39	(see Table 3)
Blue	 40-44	(see Table 3)
Navy	 45-49	(see Table 3)
Purple	 50-54	(see Table 3)
Maroon	 55-59	(see Table 3)
Red	 60-64	(see Table 3)
Orange	 65-69	(see Table 3)
Yellow	 70-74	(see Table 3)
Lime	 75-79	(see Table 3)
Green	 80-84	(see Table 3)
Emerald	 85-89	(see Table 3)

Masterplan computer model

CRITERION 1: Wharf/Dredger noise sources only, A-weighted broadband

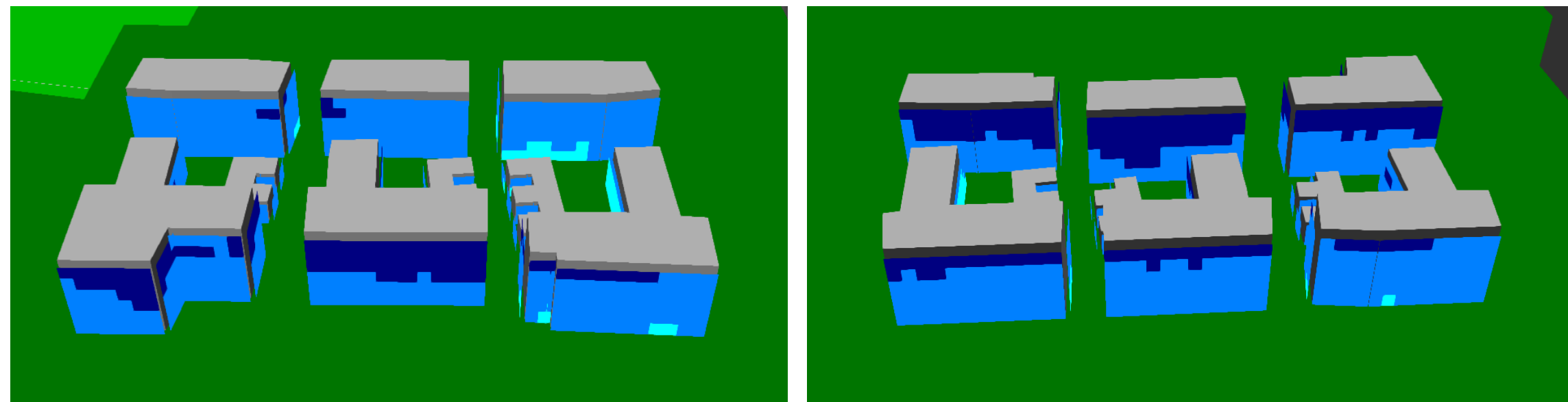


Figure 11 Masterplan model, Criterion 1 - Looking north (left) and south (right)

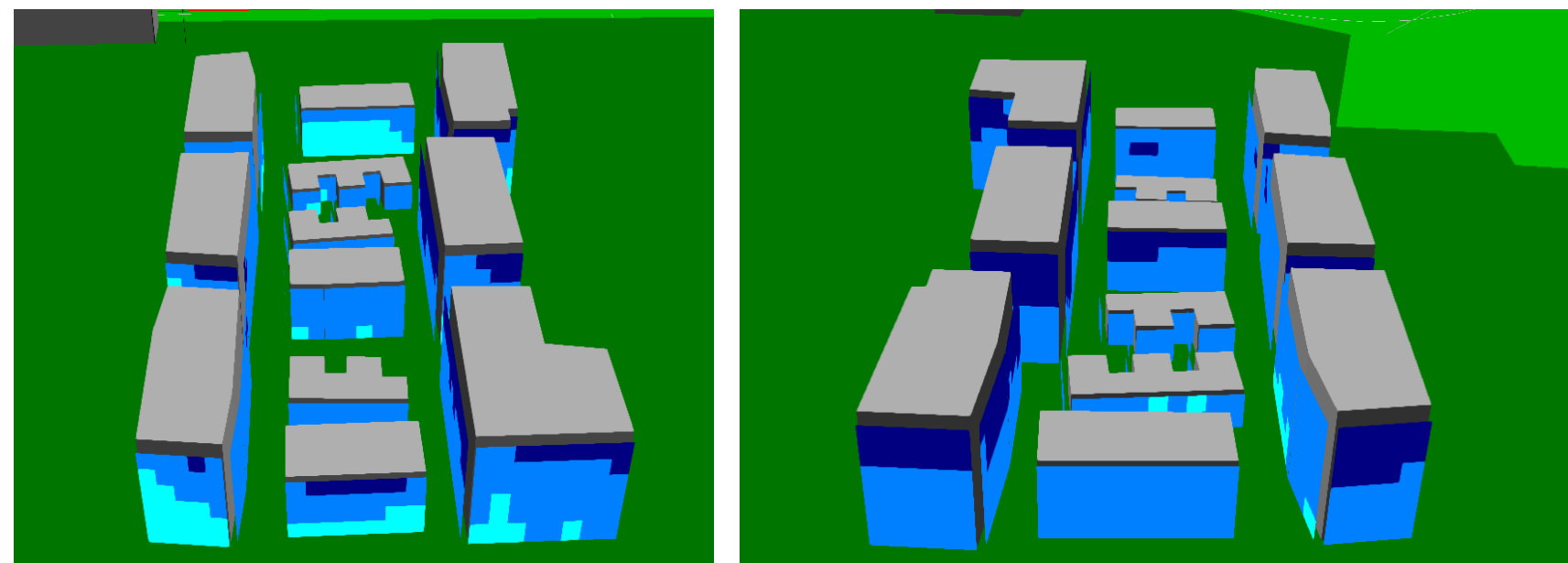


Figure 12 Masterplan model, Criterion 1 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Requirement
Grey	 ≤ 34	Windows can open
Cyan	 35-39	Windows can open
Blue	 40-44	Windows can open
Navy	 45-49	Habitable room windows fixed shut*
Purple	 50-54	Habitable room windows fixed shut*
Maroon	 55-59	Habitable room windows fixed shut

* Unless localised screening is provided.

Masterplan computer model

CRITERION 2: Dredger noise sources only, A-weighted broadband, 63 Hz dominates

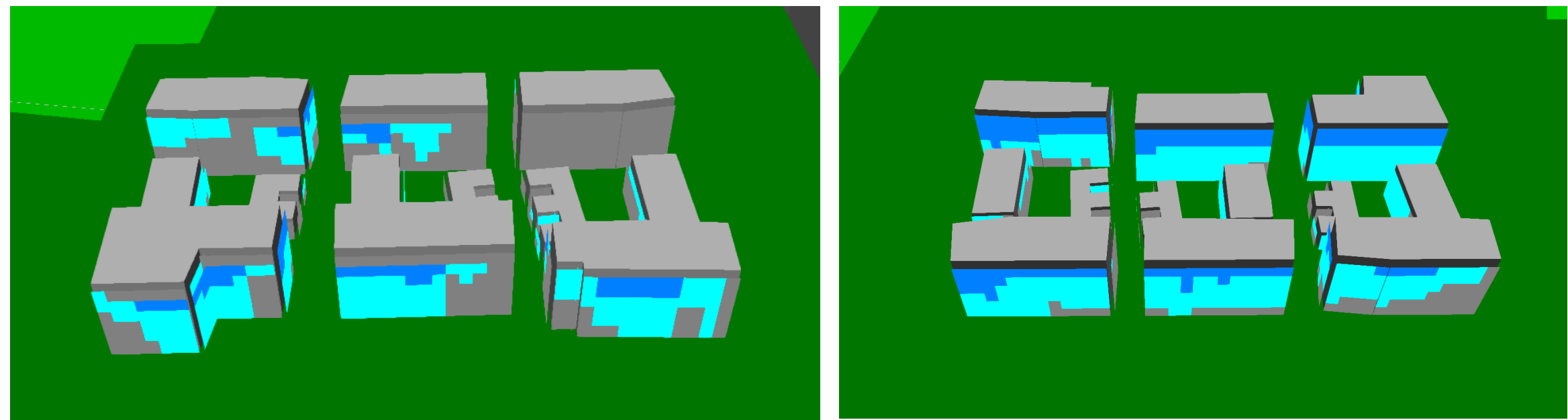


Figure 13 Masterplan model, Criterion 2 - Looking north (left) and south (right)

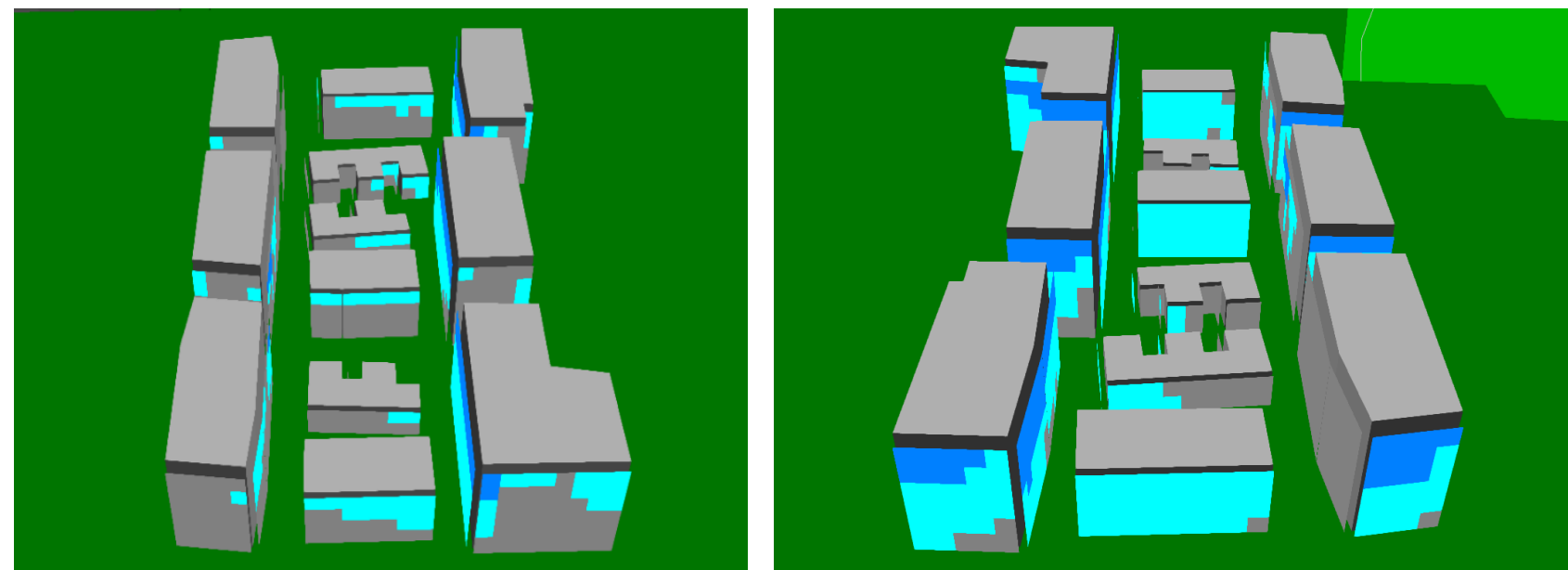


Figure 14 Masterplan model, Criterion 2 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Window Recommendation (See Table 1 for sound insulation)
Grey	 ≤ 34	Windows can open if Criterion 1 met
Cyan	 35-39	Windows open partially if Criterion 1 met
Blue	 40-44	Habitable room windows fixed shut*
Navy	 45-49	All windows fixed shut*
Purple	 50-54	All windows fixed shut
Maroon	 55-59	All windows fixed shut

* Unless localised screening is provided.

Masterplan computer model

CRITERION 3: All noise sources, NIGHT, A-weighted broadband facade noise levels

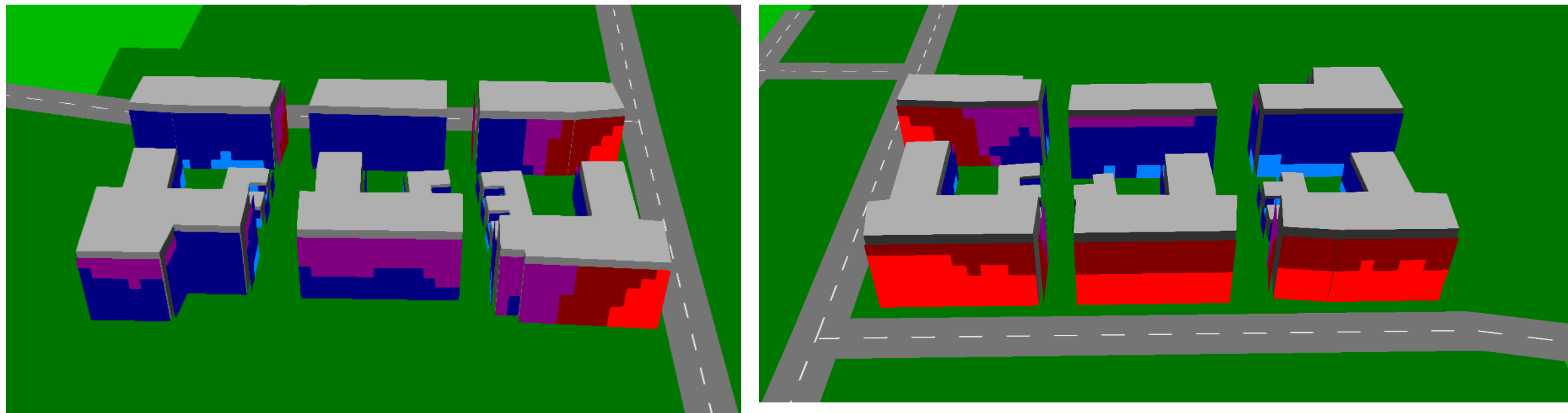


Figure 15 Masterplan model, Criterion 3 - Looking north (left) and south (right)

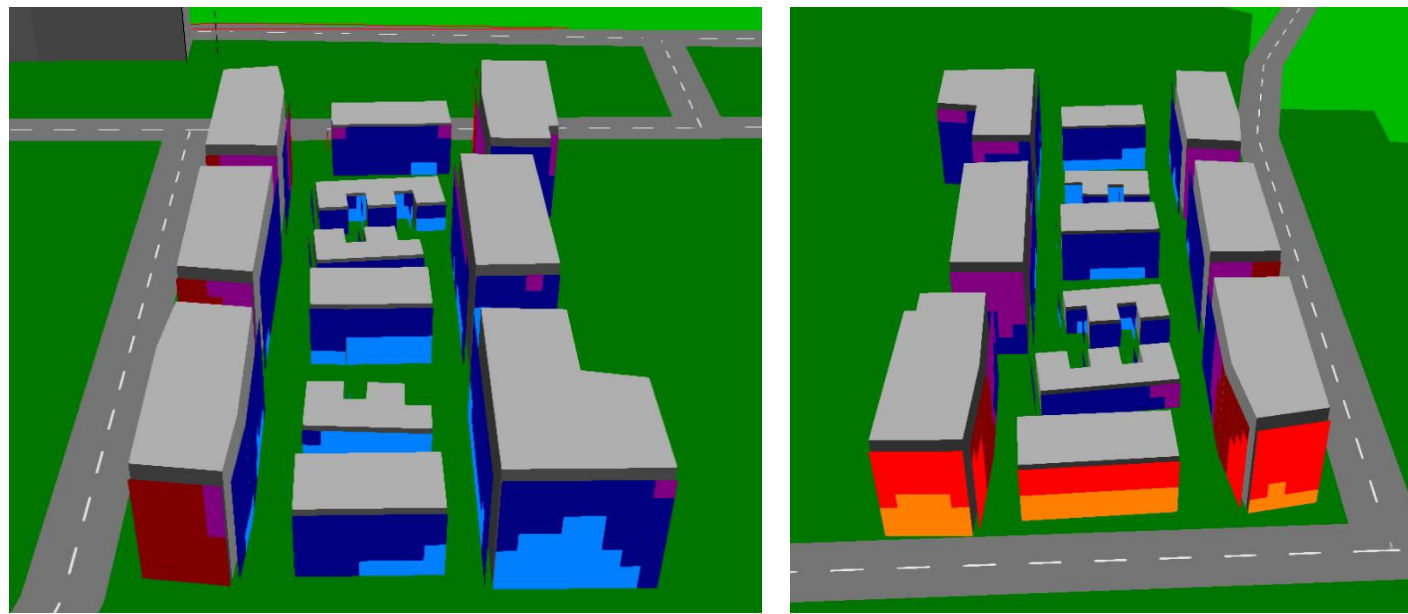


Figure 16 Masterplan model, Criterion 3 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Recommendation
Grey	 ≤ 34	(see Table 2)
Cyan	 35-39	(see Table 2)
Blue	 40-44	(see Table 2)
Navy	 45-49	(see Table 2)
Purple	 50-54	(see Table 2)
Maroon	 55-59	(see Table 2)
Red	 60-64	(see Table 2)
Orange	 65-69	(see Table 2)
Yellow	 70-74	(see Table 2)
Lime	 75-79	(see Table 2)
Green	 80-84	(see Table 2)
Emerald	 85-89	(see Table 2)

Masterplan computer model

CRITERIA 4 & 5: All noise sources, DAY, A-weighted broadband facade noise levels

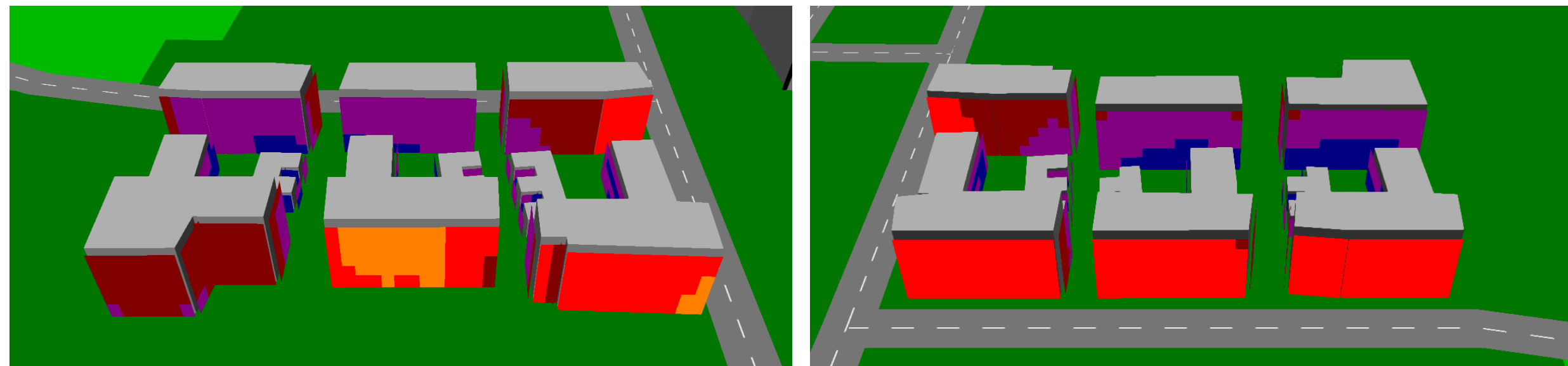


Figure 17 Masterplan model, Criteria 4 & 5 - Looking north (left) and south (right)

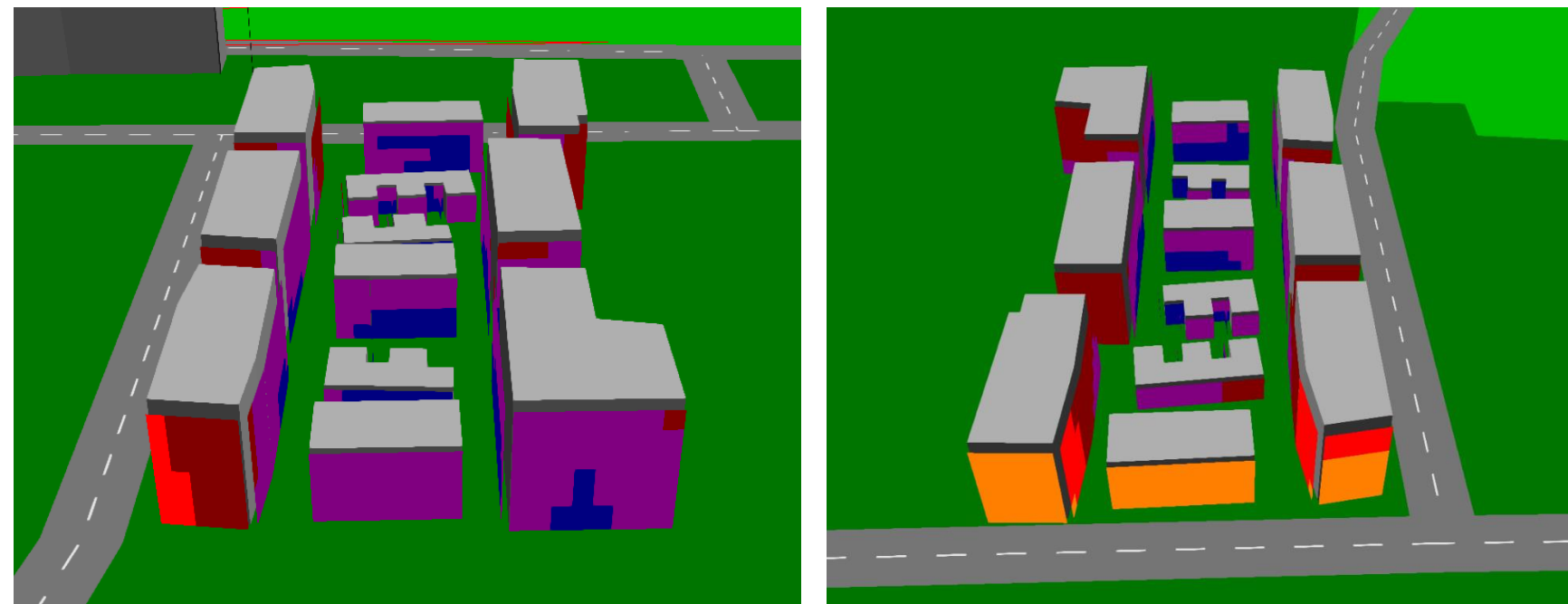


Figure 18 Masterplan model, Criteria 4 & 5 - Looking east (left) and west (right)

Colour	A-Weighted Broadband Facade Noise Level (dB)	Recommendation
Grey	 ≤ 34	(see Table 3)
Cyan	 35-39	(see Table 3)
Blue	 40-44	(see Table 3)
Navy	 45-49	(see Table 3)
Purple	 50-54	(see Table 3)
Maroon	 55-59	(see Table 3)
Red	 60-64	(see Table 3)
Orange	 65-69	(see Table 3)
Yellow	 70-74	(see Table 3)
Lime	 75-79	(see Table 3)
Green	 80-84	(see Table 3)
Emerald	 85-89	(see Table 3)

Table 1 Colour KEY to Figure 5 to Figure 6 and Figure 13 to Figure 14 – (Facade noise levels from dredgers only, 63 Hz Octave-Band, A-weighted)

CRITERION 2	HABITABLE ROOMS	NON-HABITABLE ROOMS
	($L_{Aeq,63Hz}$ 24 dB recommended limit)	($L_{Aeq,63Hz}$ 29 dB recommended limit)
Facade noise level	Sound Insulation	Sound Insulation
 Red ($L_{Aeq,63Hz}$ 60-64 dB)	D_{63Hz} 40 dB	D_{63Hz} 35 dB
 Maroon ($L_{Aeq,63Hz}$ 55-59 dB)	D_{63Hz} 35 dB	D_{63Hz} 30 dB
 Purple ($L_{Aeq,63Hz}$ 50-54 dB)	D_{63Hz} 30 dB	D_{63Hz} 25 dB
 Navy ($L_{Aeq,63Hz}$ 45-49 dB)	D_{63Hz} 25 dB	D_{63Hz} 20 dB
 Blue ($L_{Aeq,63Hz}$ 40-44 dB)	D_{63Hz} 20 dB	D_{63Hz} 15 dB
 Cyan ($L_{Aeq,63Hz}$ 35-39 dB)	D_{63Hz} 15 dB	D_{63Hz} 15 dB
 Grey ($L_{Aeq,63Hz}$ 0-34 dB)	D_{63Hz} 15 dB	D_{63Hz} 15 dB

NOTES:

- 1) The more onerous criterion from this table or from Table 2 or Table 3 shall control the specification.
- 2) Low frequency noise data for glazing has been estimated as glazing data do not normally extend down to 63 Hz. Specific testing of the agreed glazing configurations will be required.
- 3) Where non-habitable rooms open on to habitable rooms through wall openings or open doors, the resulting noise level in the habitable room shall be assessed before allowing the window to open.
- 4) Sound insulation values are indicative. Specific performances for each area in Buildings 302, 303 & 304 are given in Section 3.2.

Table 2 Colour KEY Figure 7 to Figure 8 and Figure 15 to Figure 16– (NIGHT-TIME Sound insulation, example constructions / ventilation strategies)

CRITERION 3 & 6	HABITABLE ROOMS	NON-HABITABLE ROOMS
	(L_{Aeq} 30 dB at night as per BS 8233)	(L_{Aeq} 35 dB at night recommended)
Facade noise level	Sound Insulation	Sound Insulation
 Lime (L_{Aeq} 75-79 dB)	$D_w + C_{tr}$ 50 dB	$D_w + C_{tr}$ 45 dB
 Yellow (L_{Aeq} 70-74 dB)	$D_w + C_{tr}$ 45 dB	$D_w + C_{tr}$ 40 dB
 Orange (L_{Aeq} 65-69 dB)	$D_w + C_{tr}$ 40 dB	$D_w + C_{tr}$ 35 dB
 Red (L_{Aeq} 60-64 dB)	$D_w + C_{tr}$ 35 dB	$D_w + C_{tr}$ 30 dB
 Maroon (L_{Aeq} 55-59 dB)	$D_w + C_{tr}$ 30 dB	$D_w + C_{tr}$ 25 dB
 Purple (L_{Aeq} 50-54 dB)	$D_w + C_{tr}$ 25 dB	$D_w + C_{tr}$ 20 dB
 Navy (L_{Aeq} 45-49 dB)	$D_w + C_{tr}$ 20 dB	$D_w + C_{tr}$ 15 dB
 Blue  Grey (L_{Aeq} 0-44 dB)	$D_w + C_{tr}$ 15 dB	$D_w + C_{tr}$ 15 dB

NOTES:

- 1) The more onerous sound insulation criterion from this table or from Table 1 or Table 3 shall control the specification.
- 2) As per Criterion 6, where maximum internal noise levels in bedrooms are expected to regularly exceed L_{AFmax} 45 dB (see Section 4.2), these sound insulation performances will need to be increased accordingly.
- 3) See Figure 3 to Figure 6 and Figure 11 to Figure 14 to determine if windows must be fixed closed regardless of the specification above.
- 4) Sound insulation values are indicative. Specific performances for each area in Buildings 302, 303 & 304 are given in Section 3.2.

Table 3 Colour KEY to Figure 9 to Figure 10 and Figure 17 to Figure 18 – (DAYTIME Sound insulation, example constructions / ventilation strategies)

CRITERIA 4	BEDROOMS/ NURSERY/ COMMUNITY CTR (L_{Aeq} 35 dB during the day)	OTHER HABITABLE ROOMS / OFFICES (L_{Aeq} 40 dB during the day)	NON-HABITABLE ROOMS / RETAIL (L_{Aeq} 45 dB during the day)
Facade noise level	Sound Insulation	Sound Insulation	Sound Insulation
 Lime (L_{Aeq} 75-79 dB)	$D_w + C_{tr}$ 45 dB	$D_w + C_{tr}$ 40 dB	$D_w + C_{tr}$ 35 dB
 Yellow (L_{Aeq} 70-74 dB)	$D_w + C_{tr}$ 40 dB	$D_w + C_{tr}$ 35 dB	$D_w + C_{tr}$ 30 dB
 Orange (L_{Aeq} 65-69 dB)	$D_w + C_{tr}$ 35 dB	$D_w + C_{tr}$ 30 dB	$D_w + C_{tr}$ 25 dB
 Red (L_{Aeq} 60-64 dB)	$D_w + C_{tr}$ 30 dB	$D_w + C_{tr}$ 25 dB	$D_w + C_{tr}$ 20 dB
 Maroon (L_{Aeq} 55-59 dB)	$D_w + C_{tr}$ 25 dB	$D_w + C_{tr}$ 20 dB	$D_w + C_{tr}$ 15 dB
 Purple (L_{Aeq} 50-54 dB)	$D_w + C_{tr}$ 20 dB	$D_w + C_{tr}$ 15 dB	$D_w + C_{tr}$ 15 dB
 Navy			
 Blue	$D_w + C_{tr}$ 15 dB	$D_w + C_{tr}$ 15 dB	$D_w + C_{tr}$ 15 dB
 Grey (L_{Aeq} 0-49 dB)			

NOTES:

- 1) The more onerous sound insulation criterion from this table or from Table 1 or Table 2 shall control the specification.
- 2) See Figure 3 to Figure 6 and Figure 11 to Figure 14 to determine if windows must be fixed closed regardless of the specification above.
- 3) Sound insulation values are indicative. Specific performances for each area in Buildings 302, 303 & 304 are given in Section 3.2.

3.2 Facade design strategy

In line with the strategies for sustainability and the environment, all dwellings will be provided with constant low velocity mechanical ventilation with heat recovery (LVMVHR) or similar. Even where dwellings can be naturally ventilated without excessive noise intrusion, these systems are desirable to maintain air quality and to limit heat loss in wintertime.

A general application of LVMVHR provides attenuated ventilation to all dwellings so that the provision of fresh air does not depend upon opening windows. Nevertheless, opening windows will be required to provide natural purge ventilation to the spaces. In the majority of cases it will be possible to open windows without experiencing excessive noise levels. However, in some cases it will not be possible to do so and the proposed solutions to these conditions in Buildings 302, 303 & 304 fall into three categories:

- Balconies with windows / doors where the predicted facade noise levels are above the requirements of Criterion 1 will be semi-enclosed to allow the noise levels at the facade within to be low enough to allow the windows or doors to open;
- Unscreened windows experiencing noise levels above the requirements of Criterion 1 will be fixed shut and purge ventilation will be provided by a nearby window / balcony with compliant noise levels;
- Additional screening will be provided to windows, such as with 'Juliet' arrangements, where the facade noise levels exceed the requirements of Criterion 1 and where purge ventilation cannot be provided by a nearby window / balcony with compliant noise levels.

This strategy is described in detail in Section 4.

The open area of the balcony / screens can be limited using solid balustrades and / or full height screens where necessary. Windows and doors which open onto the balcony area should be located behind the screens so as not to be in the direct sound field. As the allowable percentage open area reduces, the architectural arrangement becomes less like a balcony and more like an enclosed winter garden with opening windows.

The maximum allowable open areas for windows / doors and whether they are allowed to open are set out in the accompanying reports:

- 17470-R04-B *Building 302 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- 17470-R05-B *Building 303 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- 17470-R06-B *Building 304 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;

The revision of 17470-R04, R05 & R06 given above is the latest at the time the present report was written. Should the above report be revised, the references given herein will automatically apply to the latest version of 17470-R04, R05 & R06.

3.2.1 Elevation mark-ups

There are areas of Buildings 302, 303 & 304 where it is advised that windows/doors are fixed closed, due to either Criterion 1 or Criterion 2 being exceeded, these are marked blue in Figure 19 to Figure 31. Windows are marked green in Figure 19 to Figure 31 where Juliet screening has been utilised to enable windows to open. Further information on the amount of screening required is detailed in reports:

- 17470-R04-B *Building 302 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- 17470-R05-B *Building 303 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- 17470-R06-B *Building 304 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;

Where the requirements of Criterion 1 are not exceeded and the recommendations of Criterion 2 are exceeded by less than 2 dB, a relaxation may be possible depending on facade noise level and room condition. This is detailed further in the report mentioned above (17470-302, 303 & 304).



Figure 19 Buildings A-E-J north elevation



Figure 20 Buildings A-E-J south elevation



Figure 21 Buildings D-H-M north elevation



Figure 22 Buildings D-H-M south elevation



Figure 23 Buildings A-B-D west elevation

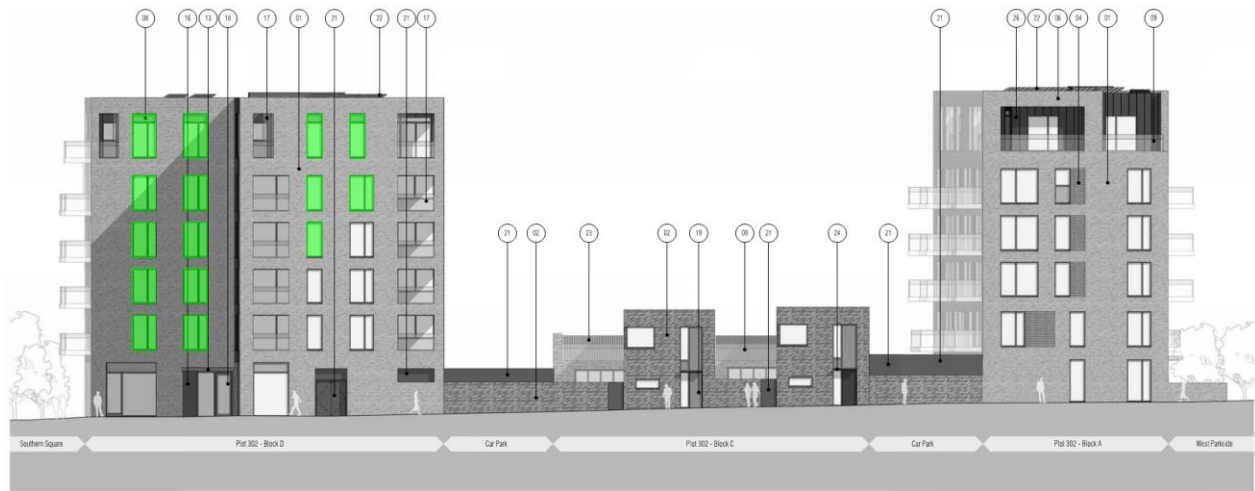


Figure 24 Buildings D-C-A east elevation

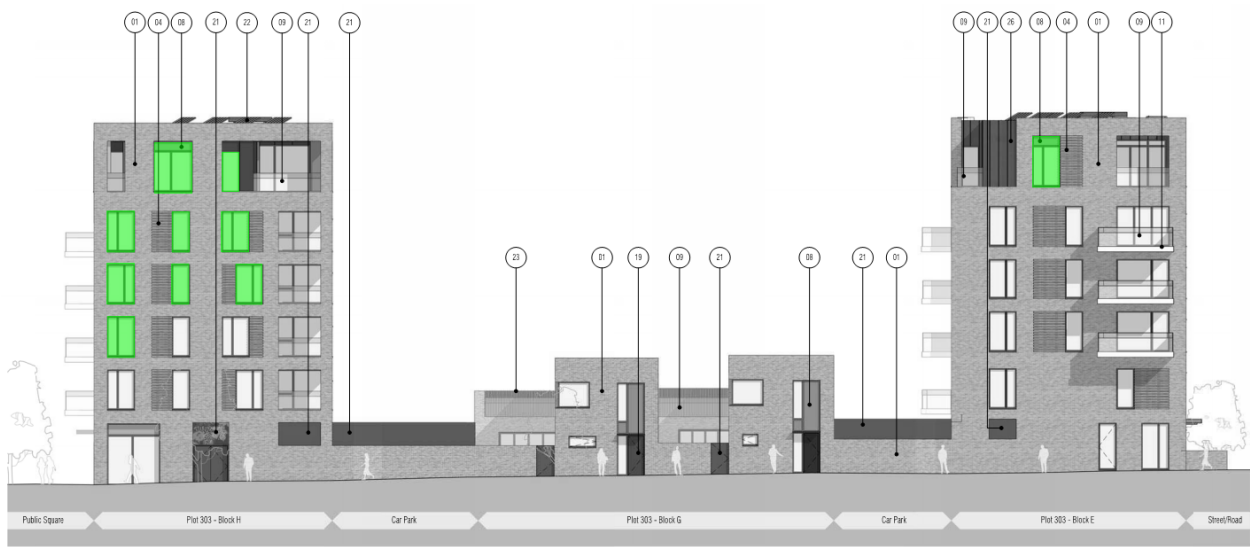


Figure 26 Buildings H-G-E east elevation



Figure 25 Buildings E-F-H west elevation

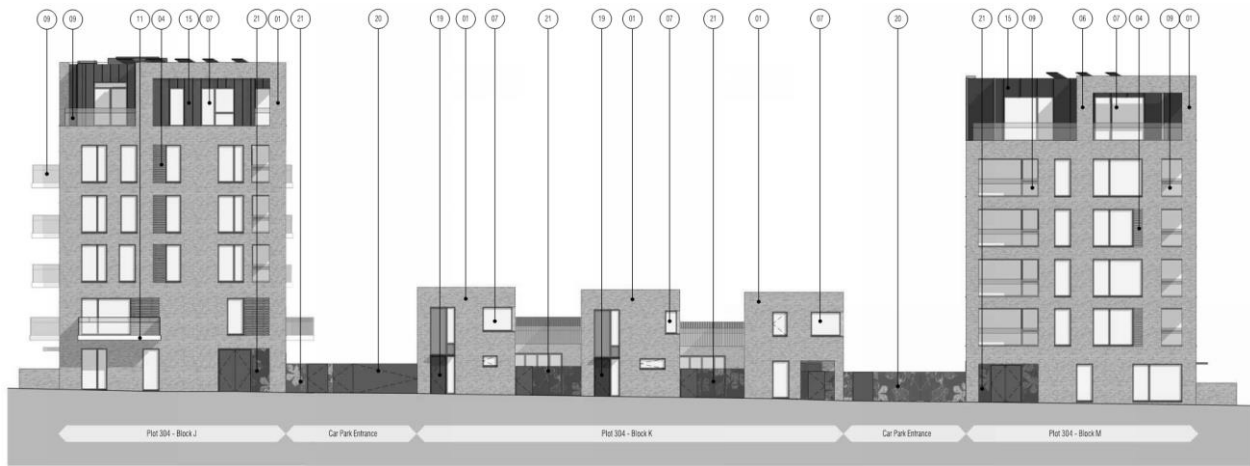


Figure 27 Buildings H-K-M west elevation



Figure 28 Buildings M-L-J east elevation

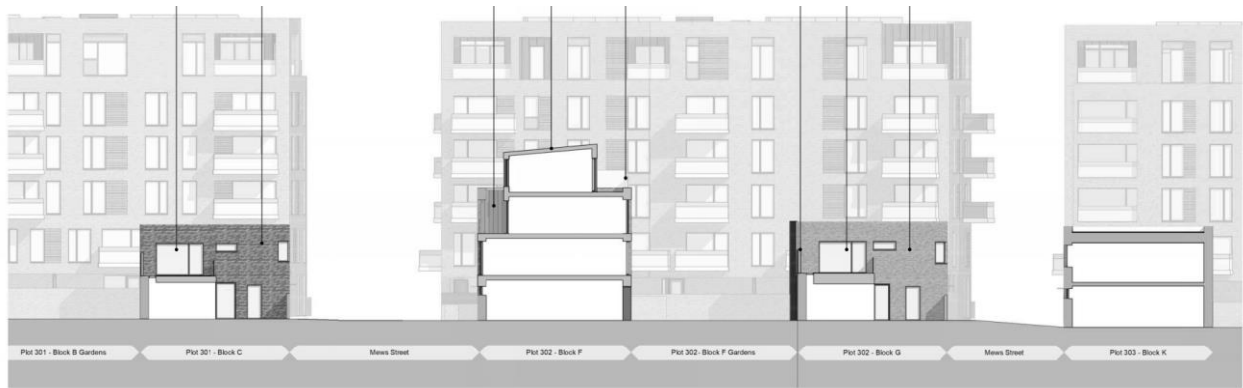


Figure 29 Buildings C-K-G-K section AA

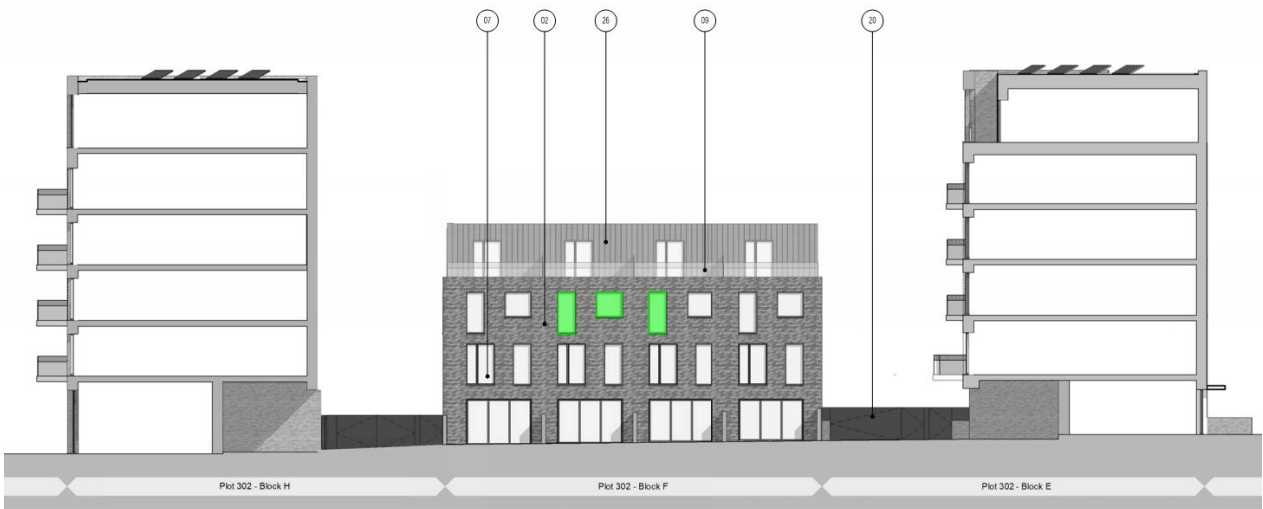


Figure 30 Buildings H-F-B section BB

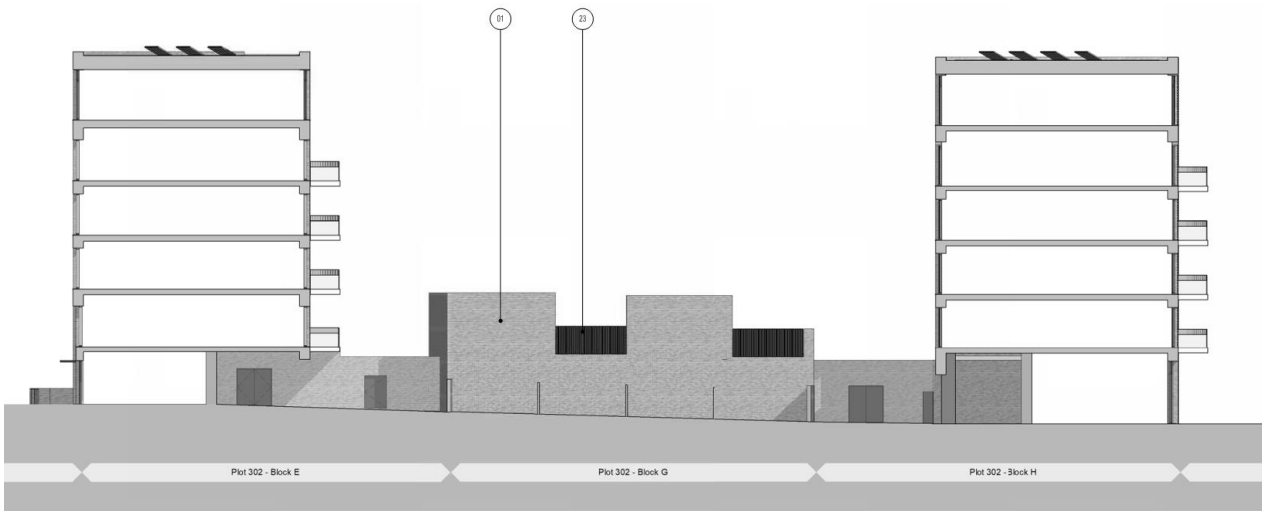


Figure 31 Buildings E-G-H section CC

Facade sound insulation performance

The required sound insulation performances for the various facades of Buildings 302, 303 & 304, along with suitable example constructions, are set out in the in the accompanying reports:

- Sandy Brown’s report 17470-R04-B *Building 302 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- Sandy Brown’s report 17470-R05-B *Building 303 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;
- Sandy Brown’s report 17470-R06-B *Building 304 facade sound insulation mark-ups – balconies and windows* dated 17 April 2018;

3.2.2 Laboratory tests

All windows and doors are required to be tested in laboratory conditions. The tests shall include results down to 50 Hz, 63 Hz and 80 Hz one-third octave bands.

All testing shall have been conducted on samples representative of the configuration of the cladding elements, including glazed and blank panels, typical joints, mullions, transoms, mullion infill elements, transom infill elements, blinds and return air slots as appropriate.

All testing shall be carried out in accordance with the standards set out in Section 3.2.3 of this specification.

It should be noted that any individual laboratory test result that does not exceed the required performance value by at least 2 dB does not provide a reliable indication that the required performance will in fact be achieved, because of the level of reproducibility of such acoustic tests.

3.2.3 Standards

Sound insulation tests shall be undertaken in accordance with:

- BS EN ISO 10140 Parts 1 to 5 *Laboratory measurement of sound insulation of building elements*;
- BS EN ISO 16283-1 *Field measurement of sound insulation in buildings and of building elements Part 1: Airborne sound insulation*,

and rated in accordance with BS EN ISO 717-1 *Rating of sound insulation in buildings and of building elements. Part 1: Airborne sound insulation*.

3.2.4 Facade sound insulation

The Weighted Sound Reduction Index and Spectrum Adapted Weighted Sound Reduction Index ($R_w + C_{tr}$) performances stated herein are defined according to BS EN ISO 10140.

Submissions shall comprise or be based on values obtained in tests carried out in a UKAS accredited laboratory in accordance with this standard.

3.3 Ventilation strategy and acoustics

Due to the urban nature of the site, the sound insulation of the building envelope relies on the windows being closed to provide the needed performance.

As such, background ventilation of the buildings will be provided by a mechanical ventilation with heat recovery system which will have the added benefit of providing exhaust for the kitchen and toilets as well as incorporating reuse of heat. See the mechanical services engineer’s information on the configuration of the ventilation strategy. However, natural purge ventilation is to be provided by opening of windows.

Figure 3 to Figure 6 and Figure 11 to Figure 14 show generally where windows are to be fixed closed and mechanical ventilation is required. These are summarised in reports 17470-R04, R05 & R06.

3.4 Flanking sound transmission at facade

Horizontal flanking sound insulation refers to the situation where partitions connect to the building envelope to form two rooms or spaces and sound is transmitted from one space to another via the cladding.

Vertical flanking sound insulation refers to the transmission of sound from one space to another that is immediately above or below via the cladding, including the slab edge detail.

Where the facade abuts internal partitions or slabs, the sound insulation it provides between horizontally adjacent rooms needs to be such that it does not undermine the sound insulation provided by internal walls.

The sound insulation performance requirements for the cladding are given in Table 4. These are based on the requirement to achieve a sound insulation performance 5 dB higher than Part E of the Building Regulations for party walls and floors, and 10 dB higher for walls and floors shared with noisier spaces.

Table 4 Flanking sound insulation performance

Location	Minimum flanking sound insulation ($D_{n,f,w} + C_{tr}$, dB)
Residence to residence	55
Residence to commercial/retail or plant	60
Residence to other areas	55

To meet this requirement, each facade in the project will require two layers of plasterboard be furred on metal studs on the inside from slab to slab and between party walls after they are constructed, as referenced in Table 2 and Table 3.

3.5 Self-noise

Noise from thermal movement and other causes will need to be controlled by appropriate specification clauses. Suitable flexible elements and movement joints are likely to be required which major cladding contractors should be familiar with.

Consideration should be given to potential resonance effects as a result of airflow across any external architectural grilles, louvres, canopies, and the like. It is recommended that elements with an unusual shape are tested in a wind tunnel at varying angles or orientation.

3.6 Rainfall noise

Noise from rainfall, particularly on lightweight horizontal or near horizontal sections of cladding, will need to be controlled by appropriate specification clauses.

Where relevant, compressed mineral fibre insulation or a suitable damping material on the inside of lightweight cladding panels would be a typical means of controlling this.

4 Example acoustics assessments

The following sections give examples of how the noise levels determined by the computer modelling are assessed against the acoustic criteria in order to develop the design of the development in terms of materials and construction types.

4.1 General notes on example calculation assessments (following pages)

The following pages show the example process by which each facade is assessed to determine the construction configuration required to meet the internal and external noise level criteria established.

In the example, a plan of the apartment under consideration is shown, followed by plans cut through the commensurate floor from each of the computer models (for the different noise sources and times of the day) in the ‘At occupation’ configuration. The facade noise levels for each are shown every 3 m, circled.

The example assessments are included in this report to demonstrate how the principles are applied to the different buildings across the development.

The models for the overall masterplan were also consulted to capture instances where noise levels may increase once later phases are built. It can be seen comparing these data that noise from traffic is louder for the buildings in Parcel 1 before the rest of the Parcels are built, as would be expected since there are no buildings between the parcel and the main roads. In addition, in some areas, wharf noise and traffic noise increases once the rest of the buildings are built, as there will be reflections off the as yet unbuilt buildings.

Facade sound level spectra for the calculations are also taken from the computer models. However, the computer models calculate traffic noise using single number values (CRTN) not spectral data, so the spectral output only includes the industrial noise sources. Therefore, a representative traffic noise spectrum (from BS 8233) has been used and adjusted taking into account the industrial noise sources to match the resulting A-weighted value. On eastern facing facades, this is not an issue as the industrial sources from the Aggregates Zone are the dominant noise source.

These assessments are based on the predicted L_{Aeq} facade noise levels. The effect of maximum noise levels is discussed in section 4.2.

For each assessment, the worst case condition determines which construction should be used based on the performances in the relevant tables (Table 1, Table 2 and Table 3). In many cases, the low frequency criterion is controlling, thus a more robust construction is used in the calculation.

For each sound insulation calculation, the reverberation time assumed is slightly higher (more conservative) than the 0.5 seconds used in standard practice for residences. This is to account for the modern trend of homes with hard surface floors and more minimalist furnishings, which affects the resulting noise level in habitable rooms. This means the sound insulation of the facade must increase further to account for this. This effect is more pronounced in living rooms than bedrooms. The

reverberation time used in the calculations has also been slightly increased at lower frequencies, again to provide a more conservative assessment. The reverberation times used are as follows.

Table 5 Reverberation times assumed in calculations

Room type	Reverberation time (T , seconds) at octave band centre frequency (Hz)						
	63	125	250	500	1000	2000	4000
Bedrooms	0.8	0.7	0.6	0.6	0.6	0.6	0.6
Other habitable rooms	1.0	0.9	0.8	0.8	0.8	0.8	0.8

4.2 Night time maximum noise levels

An assessment of the maximum noise levels measured during unattended noise measurements at the site was performed to determine if an increase on the performance of the different building facades is required. This assessment was based on the maximum L_{AFmax} noise levels measured on site at survey locations S and U. These surveys are reported in detail in report 17470-R01-A *Environmental noise survey report*, dated 14 March 2018. The measured fast-weighted L_{AFmax} noise levels were used in the assessment instead of the slow-weighted L_{ASmax} noise levels as suggested in PPG24 – *Planning Policy Guidance: Planning and noise*, as the former provide a worst case scenario and therefore allow a more onerous assessment of the sound insulation required for the building envelope.

The maximum night time $L_{AFmax,5min}$ noise levels measured each night during the survey are presented in Table 6 and Table 8 below. This table also provides the number of times the predicted internal noise level exceeds $L_{AFmax,5min}$ 45 dB per night. The weakest element of the facade construction has been used for the assessment, where the window has been assumed to achieve at least 30 dB of sound reduction.

The measured survey data complies with guidance in the World Health Organisation *Guidelines for community noise* 1999 which suggests indoor noise levels should not exceed 45 dB L_{Amax} more than 10 to 15 times per night for good sleep. The predicted number of exceedances per night are provided in the third column of the tables below.

Table 6 Maximum $L_{AFmax,5min}$ noise levels measured between 23:00 and 07:00 at survey location S

Date	Maximum $L_{AFmax,5min}$ measured between 23:00 and 07:00 (dB)	Number of predicted internal noise exceedances $L_{AFmax,5min} > 45$ dB per night
Monday 03/07/2017	79	2
Tuesday 04/07/2017	75	1
Wednesday 05/07/2017	75	1
Thursday 06/07/2017	76	2
Friday 07/07/2017	77	2
Saturday 08/07/2017	76	1
Sunday 09/07/2017	74	0
Monday 10/07/2017	73	0
Tuesday 11/07/2017	75	1
Wednesday 12/07/2017	79	2
Thursday 13/07/2017	83	1
Friday 14/07/2017	84	1
Saturday 15/07/2017	80	1
Sunday 16/07/2017	87	4

Table 7 Maximum $L_{AFmax,5min}$ noise levels measured between 23:00 and 07:00 at survey location U

Date	Maximum $L_{AFmax,5min}$ measured between 23:00 and 07:00 (dB)	Number of predicted internal noise exceedances $L_{AFmax,5min} > 45$ dB per night
Wednesday 13/12/2017	78	4
Thursday 14/12/2017	76	1
Friday 15/12/2017	81	4
Saturday 16/12/2017	80	2
Sunday 17/12/2017	75	1
Monday 18/12/2017	72	0
Tuesday 19/12/2017	73	0

4.3 Example calculation

An example facade sound insulation calculation is shown on the following page, the apartment shown is exposed to noise from the dredgers, noise from the wharf / aggregates site diffracting over the top of the eastern terrace buildings, and the general traffic noise in the area.

In this case, the northern facade's sound insulation is being controlled by night-time noise levels (Criterion 3) and the eastern facade by low frequency noise (Criterion 2).



Figure 32 Example floor plan

Note 1: CadnaA screenshots above are for the ‘At occupation’ (AO) condition models.

Table 8 Facade noise levels and sound insulation requirements

Room	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Target internal noise levels			Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								Daytime	Night time	Low frequency (at 63 Hz octave band)	
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	D_w+C_{tr}	D_{63Hz}	L_{Aeq} (dB)	L_{Aeq} (dB)	$L_{Aeq,63Hz}$ (dB)			
Living room, south windows	44	46	40	42	59	59	62	62	85%	45%	NO	NO	32	18	40	30	24	Purge ventilation via balcony instead
Bedroom, south window	44	46	40	41	59	59	63	62	85%	70%	YES	YES	32	17	35	30	24	Juliet screen arrangement

Table 9 Example night time facade sound insulation calculation for floor plan shown in Figure 32 (living room - south facade)

Facade area (m)	23 Element area (m)	dB at octave band centre frequency (Hz)							dBA
		63	125	250	500	1000	2000	4000	
Facade sound level (L_{eq})**		69	62	57	55	55	52	46	59
Solid clay bricks w/holes	17	27	33	37	37	45	49	52	
Insulight 4/12/4	6	20	24	20	25	35	30	35	
10log(S/A) correction		2.2	1.7	1.2	1.2	1.2	1.2	1.2	
Internal sound level (L_{eq})		47	34	31	25	15	9	5	28
A-weighting curve		-26	-16	-9	-3	0	1	1	
Internal sound level (L_{Aeq})		21	18	23	22	15	10	6	28
Required limit (L_{Aeq})		24							30

* Estimated values ** From computer models

Table 10 Example night time facade sound insulation calculation for floor plan shown in Figure 32 (bedroom - south facade)

Facade area (m)	11 Element area (m)	dB at octave band centre frequency (Hz)							dBA
		63	125	250	500	1000	2000	4000	
Facade sound level (L_{eq})**		69	62	57	55	55	52	46	59
Solid clay bricks w/holes	6	27	33	37	37	45	49	52	
Insulight 4/12/4	5	20	24	20	25	35	38	35	
10log(S/A) correction		2.2	1.6	0.9	0.9	0.9	0.9	0.9	
Internal sound level (L_{eq})		47	34	31	25	14	8	4	27
A-weighting curve		-26	-16	-9	-3	0	1	1	
Internal sound level (L_{Aeq})		20	18	22	21	14	10	5	27
Required limit (L_{Aeq})		24							30

* Estimated values ** From computer models

4.4 Strategy for marginal facade noise levels

The computer models show the facade noise level at the outer structure of each building. Where balconies and terraces extend into the building, the noise level at the inner facade will be lower and in marginal cases can allow a window or door to open where the models may show they cannot.

The principle used to calculate noise levels at the facade inside a balcony is based on a composite sound insulation calculation as used to determine the sound insulation of a facade made up of several different elements such as masonry, glazing, etc. The sound insulation performance for each element is logarithmically summed in proportion to the area the element occupies to obtain the overall sound insulation performance for the area of facade. In the case of an open balcony with a solid balustrade, this is a combination of the performance of the balustrade material and the opening, the latter having a sound insulation performance of zero.

In addition to this, once the balcony is sufficiently enclosed there is a correction for the area of the separating element and the amount of sound absorption on the balcony (assumed to be minimal). When the proportion of open area is high, the sound transmission and reverberant sound build-up do not behave the same way so this correction does not apply.

Using this method, the sound reduction for different open areas of balcony can be determined.

Table 11 and Table 12 use a typical balcony to demonstrate the calculation. The sound reductions to apply are highlighted in bold in Table 13.

The open area of the balcony can be limited using solid balustrades and / or full height screens where necessary. Windows and doors which open onto the balcony area should be located behind the screens so as not to be in the direct sound field. As the allowable percentage open area reduces, the architectural arrangement becomes less like a balcony and more like an enclosed winter garden with opening windows.

Table 11 Typical balcony dimensions

Depth	Width	Height	Volume	Surface area	Facade area
2 m	5 m	2.5 m	25 m ³	55 m ²	12.5 m ²

Table 12 Element performances

Solid element performance (R_w)	20 dB
Absorption coefficient for floor / walls (α_w)	0.1
Absorption coefficient for opening (α_w)	1.0

Table 13 Composite sound insulation performance

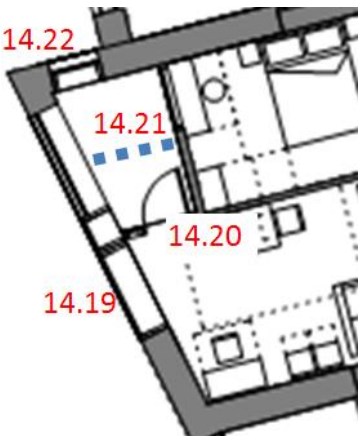
% Open area	Composite sound reduction	Correction for area and absorption (10logS/A)	Total attenuation
100	0	-1.3	1
95	0	-1.1	1
90	0	-1.0	1
85	1	-0.8	2
80	1	-0.6	2
75	1	-0.5	2
70	2	-0.3	2
65	2	-0.1	2
60	2	0.1	2
55	3	0.3	2
50	3	0.5	2
45	3	0.7	3
40	4	1.0	3
35	4	1.2	3
30	5	1.5	4
25	6	1.8	4
20	7	2.1	5
15	8	2.4	6
10	10	2.8	7
5	12	3.1	9
2.5	15	3.3	11
0	20	3.6	16

Application of strategy

Noise levels at the facades have been assessed based on the worst case situation since some noise levels increase with the introduction of buildings from later parcels of the development.

Noise levels output from the CadnaA models are shown in Section 3 for the ‘at occupation’ (AO) condition, and the later masterplan.

To assist in assessing the requirement for sealed windows and the percentage of balcony screening an example assessment is outlined below.



Ref.	Noise levels (dB)								Balcony/ shroud open area		Window/ door openable		Window/door sound insulation (dB)		Notes
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz})		All sources night (L_{Aeq})		All sources day (L_{Aeq})								
	AO	MP	AO	MP	AO	MP	AO	MP	AO	MP	$D_{eq}+C_{eq}$	D_{63Hz}			
14.19	41	41	40	40	52	50	59	58	0%	0%	NO	NO	23	16	
14.20	41	41	40	40	52	50	59	58	85%	85%	YES	YES	23	16	
14.21	42	42	42	42	52	50	59	58	45%	45%	YES	YES	25	18	
14.22	43	43	42	42	49	49	56	56	0%	0%	NO	NO	20	18	

Figure 33 Building 201, level 14 partial floor plan showing facade noise levels and assessment

The noise levels in columns 2 to 5 of the table above show the worst case noise levels for each iteration of the noise models with the different noise source scenarios. AO refers to the models with the buildings existing ‘At Occupation’ of Buildings 302, 303 & 304 in place, while MP refers to the models with the whole masterplan development in place.

Table 14 provides a key to the facade mark-up table shown in Figure 33, where the noise levels applicable to the different Criteria are highlighted.

Table 14 Key to facade mark-up table

Ref.	Noise levels (dB)								Balcony/ shroud open area	Window/ door openable		Window/door sound insulation (dB)		Notes	
	Wharf & dredgers (L_{Aeq})		Dredgers (L_{A63Hz} *)		All sources night (L_{Aeq})		All sources day (L_{Aeq})					D_w+C_{tr}	D_{63Hz}		
	AO	MP	AO	MP	AO	MP	AO	MP		AO	MP				
XX	XX	XX	XX	XX	XX	XX	XX	XX	XX%	XX%	Yes / No	Yes / No	XX	XX	

Window / door reference number
Criterion 1: $L_{Aeq} \geq 45$ dB - window fixed closed unless adequately shrouded
Criterion 2: $L_{A63Hz} \geq 40$ dB - window recommended to be closed unless shrouded; predicted levels compared to Acoustic Report to determine required D_{63Hz} for window / door
Criteria 3 & 4: predicted levels which are compared to Acoustic Report to determine required window / door sound insulation performance

* This number is approximated from the broadband models of dredger noise, which dominates at 63 Hz. The actual level at 63 Hz is equal to this number.

For the apartment shown, the required Criterion 1 facade noise level is L_{Aeq} 41 dB. No screening would be required for this apartment to allow windows to open for Criterion 1. However, the targeted Criterion 2 (L_{A63Hz} 39 dB) is exceeded by 1-3 dB for windows/door 4.19 to 4.22. The facade noise level is recommended to be 1-3 dB lower to achieve a noise level from the dredgers of L_{A63Hz} 39 dB to allow an opening window or door onto the balcony.

From Table 13, the percentage open area allowable for the outer balcony facade in front of door 14.20 and window 14.21 to reduce the noise levels by 1 - 3 dB is 85% and 45% respectively. Note, where two facade levels occur across one balcony opening, the percentage of openable area is applicable to the relevant balcony segregation (see dotted line in Figure 33). A combination of solid facade elements such as balustrades, glass or opaque panels will be provided to limit the open area.

As no balcony is proposed in front of window 14.19, and as the low frequency criterion is exceeded by 1 dB, this window is recommended to be fixed shut, and the natural ventilation provided via the adjacent balcony door 14.20 instead.

The levels at window 14.21 rely on the window at 14.22 to be sealed shut.

Appendix A

Glossary of terms

The following is a partial list of terms used in this report described.

Sound level terms

dB	Decibel – A logarithmic unit used for many acoustic values to indicate the level with respect to a reference level
dB(A)	The sound level in dB incorporating a frequency weighting (“A” weighting) which differentiates between sounds of different frequency (pitch) in a similar way to the human ear. Values in dB(A) broadly agree with people’s assessment of loudness.
Hz	Hertz – Unit of frequency, equal to one cycle per second, related to the pitch of a sound.
L_{pA}	The "A" weighted sound pressure level in dB (re 20 μ Pa). This value must be accompanied by a distance from the source.
L_{wA}	The "A" weighted sound power level in dB (re 1 picowatt).
$L_{A90,T}$	The A-weighted sound pressure level exceeded 90% of the measurement period (T) over which a noise is measured (ie, the quietest 10% of the period). When not weighted it is denoted $L_{90,T}$. This parameter is generally considered to be representative of a constant noise source, or background noise level.
$L_{Aeq,T}$	Equivalent A-weighted sound pressure level of a steady noise that has the same acoustic energy as a fluctuating noise over the measurement period (T). When not weighted it is denoted $L_{eq,T}$.
$L_{A10,T}$	The A-weighted sound pressure level exceeded for 10 % of the measurement period (T) over which a noise is measured. When not weighted it is denoted $L_{10,T}$.
$L_{Amax,T}$	The highest A-weighted sound pressure level measured in the period (T) with either fast (L_{AFmax}) or slow (L_{ASmax}) time weightings. When not weighted it is denoted L_{Fmax} or L_{Smax} .

Room acoustics terms

α_w	Weighted absorption coefficient – a quantity characterising the effectiveness of a sound absorbing surface. The proportion of sound energy absorbed is given as a number between zero (for a fully reflective surface) and one (for a fully absorptive surface).
T_{mf}	Mid-frequency reverberation time in seconds – The time in seconds taken for a sound to decay by 60dB after a sound source has been stopped, when measured at 500, 1000, and 2000 Hz and arithmetically averaged. It is based on the absorption characteristics of the room finishes.

Sound insulation terms

D_w	Weighted Level Difference in dB – A single number weighted quantity which characterizes the airborne sound insulation between rooms. It is not corrected for reverberation or element size and is therefore representative of the sound insulation heard on site. A higher numerical quantity represents a better performance.
$D_{nT,w}$	Weighted Standardized Level Difference in dB – A single number weighted quantity which characterizes the airborne sound insulation between rooms, corrected for reverberation time. A higher numerical quantity represents a better performance.
R_w	Weighted Sound Reduction Index in dB – A single number quantity which characterises the airborne sound insulation of a material or building element in the laboratory. A higher numerical quantity represents a better performance.
C_{tr}	The correction to a sound insulation quantity (such as R_w or $D_{nT,w}$) to take account of a specific sound spectrum with low frequency sound energy.
$L'_{nT,w}$	Weighted Standardised Impact Sound Pressure Level in dB – A single number weighted quantity used to characterise the impact sound insulation of floors. A lower numerical quantity represents a better performance.

Other terms

Habitable Rooms – From the London Housing Design Guide: “Habitable rooms provide the living accommodation of the dwelling. They include living room, dining room, study, home office, conservatory, bedroom, etc. They exclude the bathroom, WC, utility room, store room, and circulation space. A kitchen is not a habitable room unless it provides space for dining.” For purposes of this document, small kitchens open to living spaces will also be considered habitable.